

pop quiz Kine 1D -3

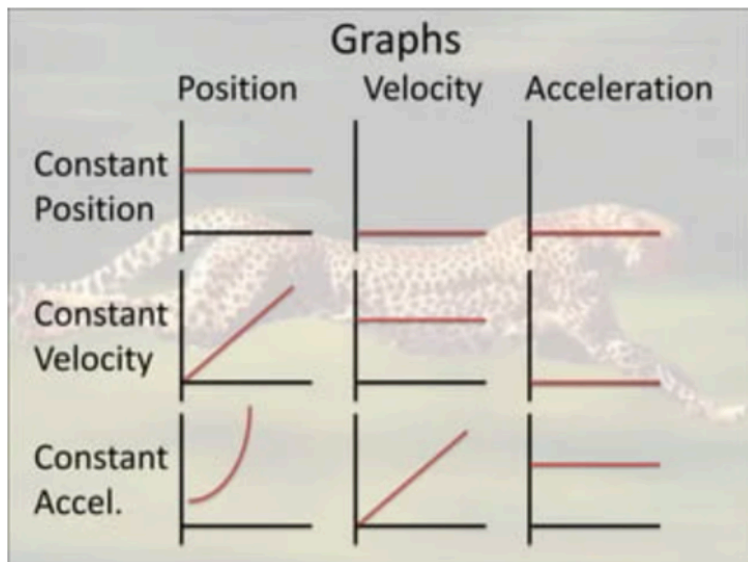
 This is a preview of the published version of the quiz

Started: Apr 29 at 7:51pm

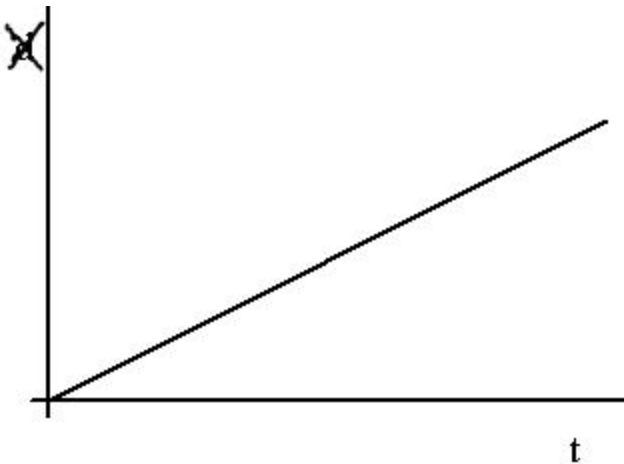
Quiz Instructions

1 attempt

hint:



Question 1 17 pts



Here is the position (on the number line) on an object as a function of time

hint: is the steepness of the line change?



its velocity is 0



its velocity is increasing



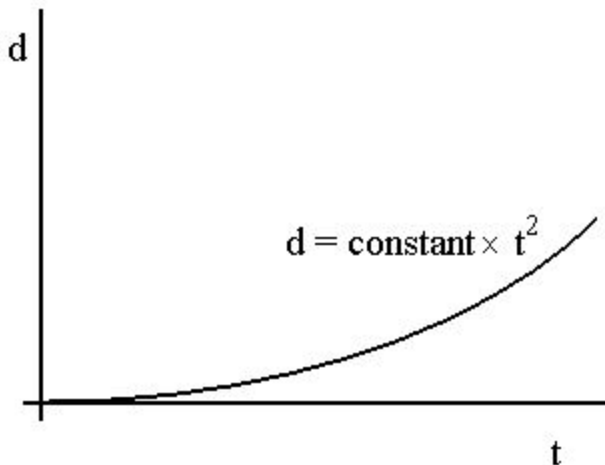
its velocity is decreasing



Its velocity is constant



Question 2 17 pts



Here is the distance from the origin versus time of an object. And the equation of the graph.

What can be said about the object moving on the number line?

hint: d is like x . The position on the number line. what kind of graph is it? visited in class



its speed is constant



speed is increasing, the acceleration is not constant



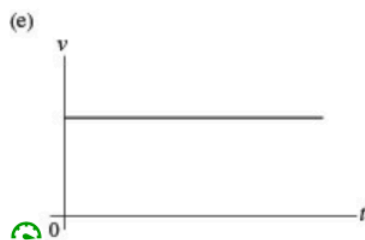
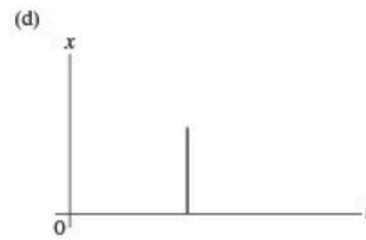
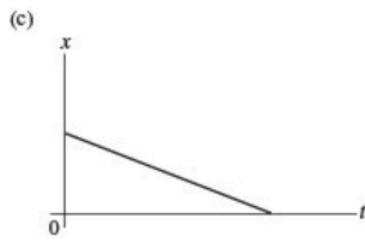
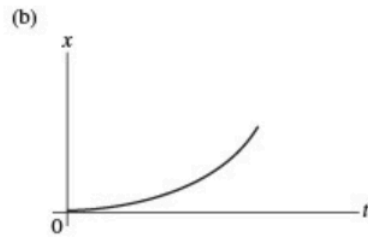
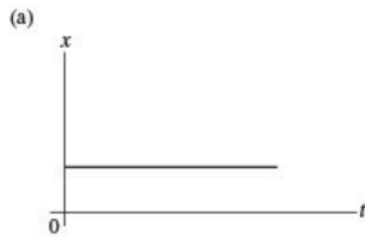
speed is 0



its speed is increasing, acceleration is constant



Question 3 17 pts



which graph is for an object at rest on the number line.

☐

d

☐

a

☐

c

☐

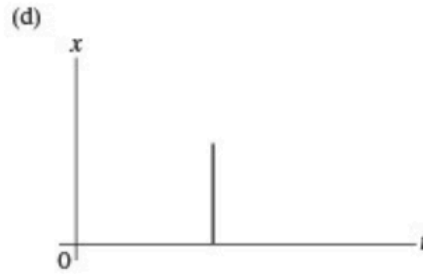
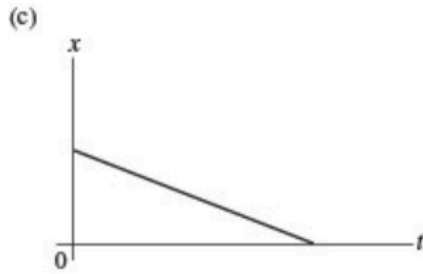
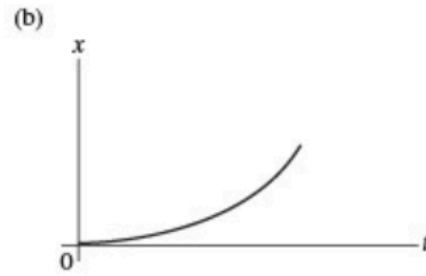
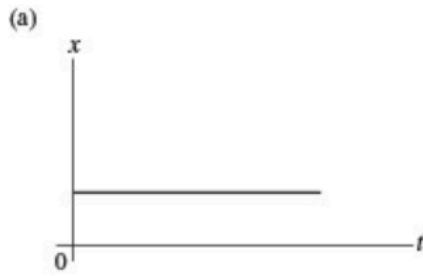
b

☐

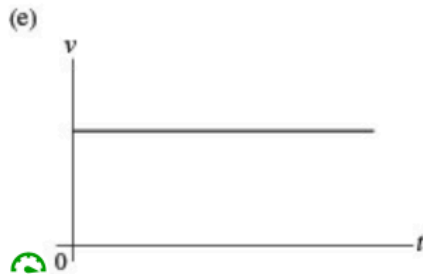
e

☐

Question 4 17 pts



following



Which graph describes an object moving @ left (negative direction) on the number line

☐

e

☐

b

☐

c

☐

a

☐

d

☒

Question 5 17 pts

An object is moving with constant non-zero velocity in the +x direction. The velocity versus time graph of this object is

☐

a parabolic curve.

☐

a horizontal straight line



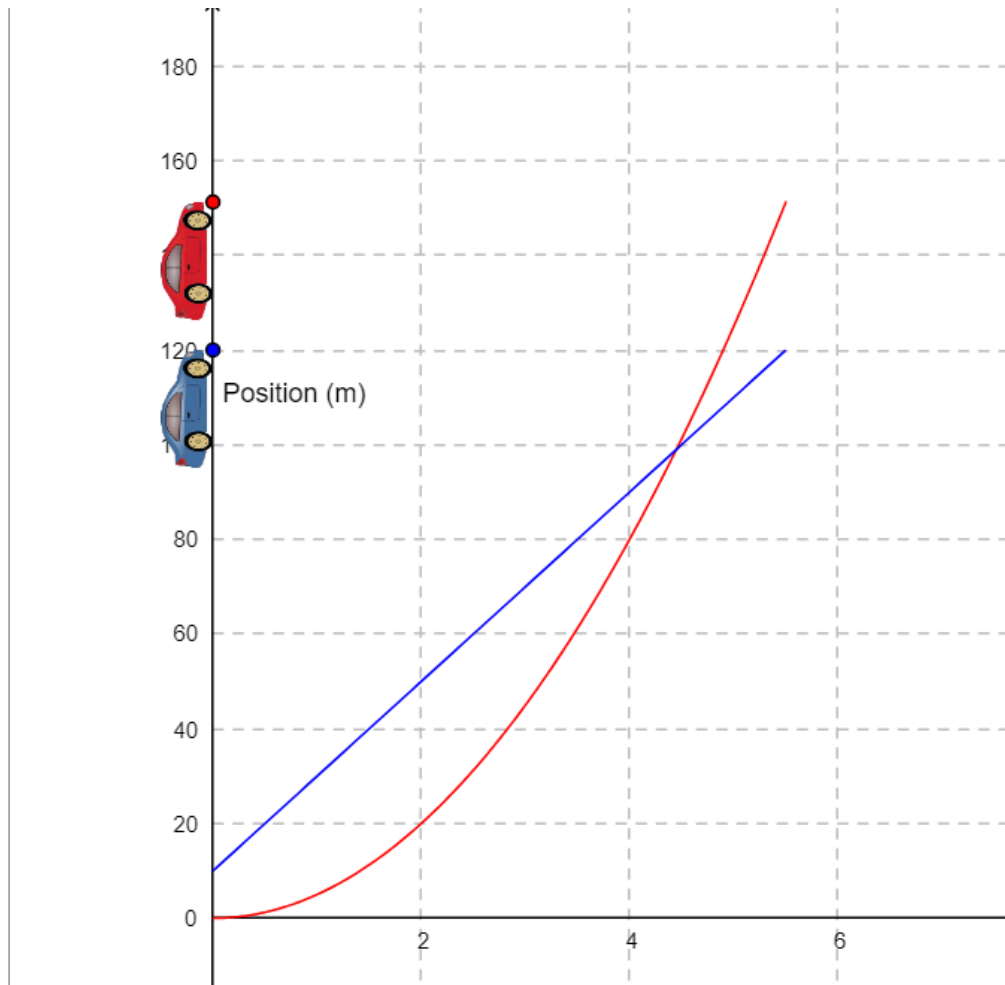
a vertical straight line.



a straight line making an angle with the time axis.



Question 6 5 pts



When the red car passes the blue car, what is x and t ? that is the position of the cars on the number line (y-axis here) and the time elapses since the race started (it starts at $(0,0)$)



(5m, 4.5s)



(120m, 5.5s)



(100m, 4.5s)



(150m, 5.5s)



Question 7 8 pts

Building on the previous question

Consider the blue car. You have the position on the number line as a function of time $x(t)$. What is the equation of motion of this car? that is what is the equation $x(t)$. So the equation of the graph

☐

$$x=20t + 10$$

☐

$$x=10t^2 + 10$$

☐

$$x=20t$$

☐

$$x=2t + 5$$



Question 8 10 pts

Building on the previous question

What is true

☐

the red car is moving at a constant velocity

☐

the red car is accelerating starting from a non zero velocity

☐

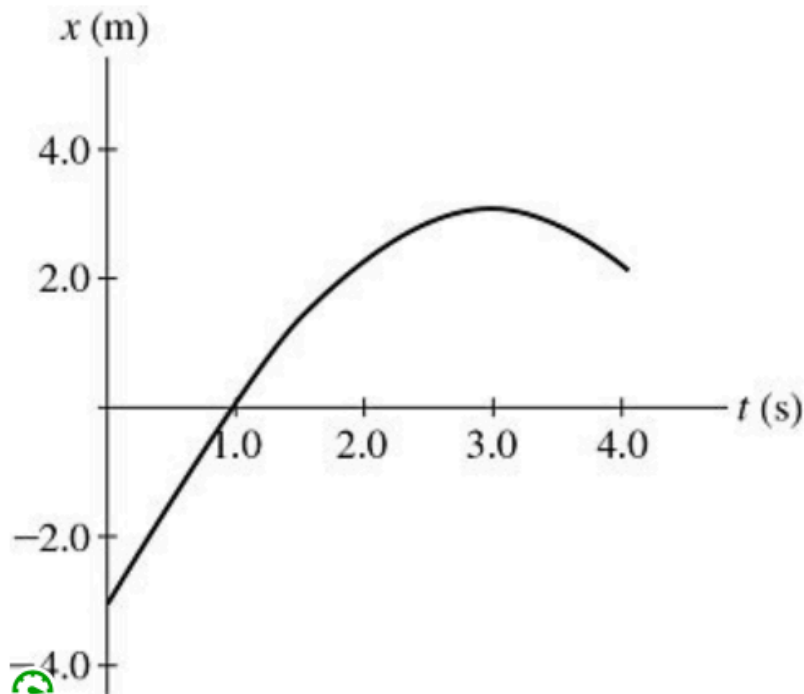
The red car is acceleratiing from rest

☐

The red car is no moving. dah



Question 9 6 pts



This graph describes the position x of a particle on the number line. It starts from position -3, reaches position 3 and reverses. At what time is the velocity of the particle 0?

(think slope)

☐

4s

☐

0s

☐

The velocity is never zero. dah

☐

2s

☐

3s

☐

1s

Not saved

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